

INTELLIGENT RESUME ANALYSIS AND RANKING

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report **“INTELLIGENT RESUME ANALYSIS AND RANKING”** is the Bonafide work of **“KISHORE PS”, “MAULYA H”, “RAMYA G”, “PRADEEP BALAN R”** submitted during the academic year 2025 – 2026, in partial fulfilment of the course **U21CS605 – PROTO STUDIO II** in Computer Science and Engineering at KPRIET.

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ABSTRACT

Intelligent Resume Analysis and Ranking System is a smart recruitment platform designed to streamline the manual resume screening process through automation and intelligent evaluation. It leverages techniques from Natural Language Processing and Machine Learning to analyze resumes, extract key details such as skills, education, experience, certifications, and project exposure, and match them with job requirements effectively. The system follows a structured and scalable architecture to perform resume parsing, semantic job description matching, and candidate ranking based on relevance, qualifications, and skill alignment. It uses intelligent scoring mechanisms to compare candidate profiles with job descriptions beyond simple keyword matching, ensuring more context-aware and meaningful evaluation. Additionally, the platform incorporates advanced features such as skill gap analysis, comparative candidate scoring, and ranking visualization to help recruiters clearly understand candidate suitability. It also supports filtering and sorting options based on experience level, technical skills, and job fit percentage. The system provides a user-friendly interface for candidates to upload resumes and view basic feedback, while offering a powerful recruiter dashboard to manage job postings, review shortlisted candidates, and analyze recruitment insights efficiently. By improving accuracy, reducing manual effort, minimizing bias, and supporting data-driven decision-making, the system significantly enhances overall recruitment efficiency and effectiveness.

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CHAPTER 1

INTRODUCTION

1.1 General

A common recruitment challenge creates a gap between traditional resume screening and modern intelligent hiring solutions. Recruiters often struggle to review large volumes of resumes, leading to time-consuming processes, inconsistent evaluations, and delayed decisions. Additionally, the process is prone to human bias and errors, making it difficult to ensure accuracy and efficiency in hiring. Furthermore, organizations lack a unified platform to effectively match candidate profiles with job requirements.

1.2 Need for the Study

There is a critical need for an intelligent resume analysis system that connects recruiters and candidates efficiently. The current hiring process needs to be transformed from a time-consuming and inconsistent method into a fast, accurate, and automated system that ensures fair evaluation and reliable candidate selection.

1.3 Objective

- **Efficient Resume Management:** To enable recruiters to upload, organize, and manage large volumes of resumes easily, reducing manual effort.
- **Automated Candidate Evaluation:** To automatically analyze resumes and rank candidates based on job requirements, ensuring accuracy and speed.
- **Fair & Transparent Screening:** To provide unbiased evaluation by focusing on skills and qualifications, ensuring consistency and transparency in hiring.
- **Flexible Criteria Configuration:** To allow recruiters to define job requirements and customize evaluation parameters for effective candidate selection.

1.4 Scope of the Project

The scope of the Intelligent Resume Analysis and Ranking System includes developing a platform for analyzing and managing resumes digitally. It provides an interface for candidates to upload resumes and receive feedback, along with a recruiter dashboard for evaluating applications, ranking candidates, and monitoring hiring insights efficiently.

CHAPTER 2

LITERATURE SURVEY

2.1. Digital Transformation in Recruitment Systems

The integration of Information and Communication Technologies (ICT) plays a crucial role in transforming traditional recruitment methods into modern intelligent systems. Traditional hiring processes—such as manual resume screening and shortlisting—are often time-consuming, inconsistent, and prone to human bias. To overcome these challenges, intelligent resume analysis systems have emerged as an effective solution, enabling automated resume parsing, skill extraction, and candidate ranking using techniques from Natural Language Processing and Machine Learning, thereby improving efficiency, accuracy, and fairness in the recruitment process (Kumar & Singh, 2022).

2.2 Review of Existing Resume Screening Systems

- **LinkedIn Recruiter:** A widely used professional hiring platform that allows recruiters to search and filter candidates based on skills, experience, and profiles. While it is effective for sourcing candidates, it relies heavily on manual filtering and does not provide deep automated ranking or detailed skill gap analysis.
- **Naukri.com:** One of the largest job portals that enables recruiters to access a vast resume database. It supports keyword-based search but lacks intelligent evaluation and semantic matching of candidate profiles with job descriptions.
- **Indeed Hiring Platform:** A global recruitment platform that provides resume access and applicant tracking features. However, its evaluation process is largely manual and does not include advanced AI-based candidate ranking or personalization.
- **Limitations of Existing Systems:** Existing recruitment platforms largely rely on keyword-based matching, which reduces the accuracy of candidate selection. They also lack advanced automation for ranking applicants based on skills and relevance. Additionally, limited use of intelligent analysis affects the efficiency and fairness of the recruitment process.

2.3 Technological Advancements in Resume Screening

To address the limitations of manual resume evaluation and traditional recruitment methods, recent developments focus on integrating advanced technologies into hiring systems. Automated resume parsing and analysis techniques are used to extract key information such as skills, education, and experience efficiently. Additionally, modern systems incorporate intelligent matching algorithms to compare candidate profiles with job requirements and support basic bias reduction mechanisms. These technological improvements enhance accuracy, speed, and consistency, making the recruitment process more efficient and scalable.

2.4 Summary and Research Gap

While existing recruitment platforms such as LinkedIn and Naukri have digitized the hiring process, they still rely heavily on manual filtering and keyword-based search, which limits accuracy and efficiency. These systems often lack advanced automation for candidate ranking, semantic matching of skills, and bias-free evaluation. The literature highlights a critical gap for a unified system that not only manages resumes digitally but also provides intelligent analysis, automated ranking, and fair assessment of candidates. The proposed Intelligent Resume Analysis and Ranking System is designed to bridge this gap by enabling automated evaluation, improved accuracy, and a more efficient and transparent recruitment process.

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

The Intelligent Resume Analysis and Ranking System is built on a modular and scalable architecture, designed with multiple components to ensure clear separation of functionality, easy maintenance, and future scalability.

- **User Management:** Handles secure registration, login, and profile management for both candidates and recruiters.
- **Resume Management:** Manages resume upload, parsing, and extraction of key information such as skills, education, and experience.
- **Evaluation System:** Responsible for analyzing resumes, matching them with job descriptions, calculating relevance scores, and generating candidate rankings.
- **Admin/Recruiter Dashboard:** Provides a backend interface for recruiters to manage job postings, view ranked candidates, and analyze recruitment insights efficiently.
- The system utilizes a secure, role-based structure where access is controlled based on user roles:
- **Candidate:** A user role that allows individuals to upload resumes, view feedback, and track application status.
- **Recruiter:** A role responsible for posting job descriptions, reviewing ranked candidates, and managing hiring processes.
- **Admin:** The administrator role with full control over the system, including user management, job postings, system configuration, and access to analytics and reports.

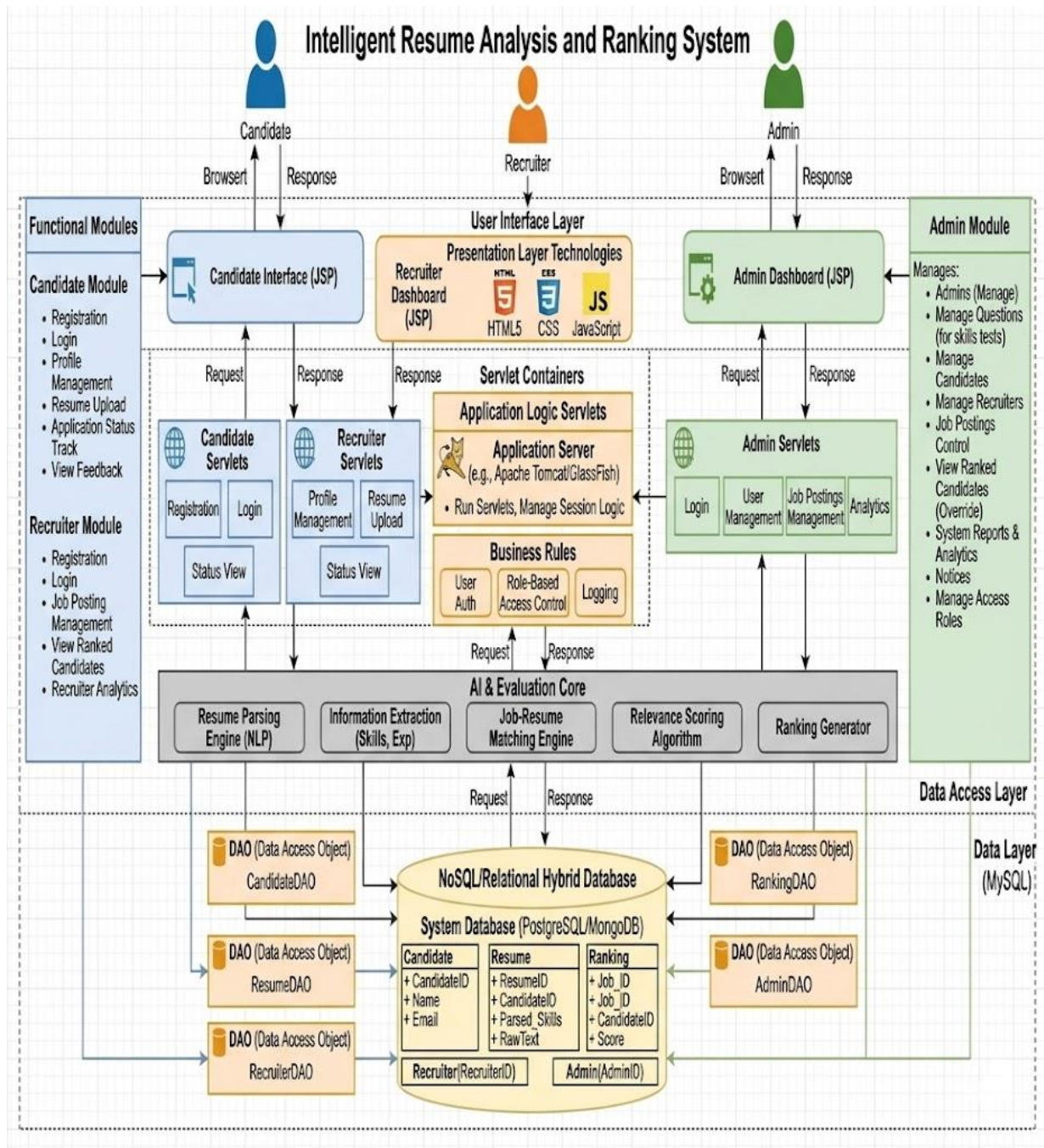


Fig.1 – System Architecture

3.2 DATAFLOW

The system is designed to provide a simple and efficient experience for candidates and recruiters through a three-step dataflow:

1. **Login & Access:** A candidate logs into the system using secure credentials and uploads their resume or accesses available job postings created by recruiters.
2. **Upload & Analysis:** The candidate's resume is processed by the system, where key information such as skills, education, and experience is extracted and matched with job requirements.
3. **Evaluate & Ranking:** The system evaluates the resume using intelligent matching techniques, generates a relevance score, and ranks candidates accordingly, allowing recruiters to view results and select suitable applicants efficiently.

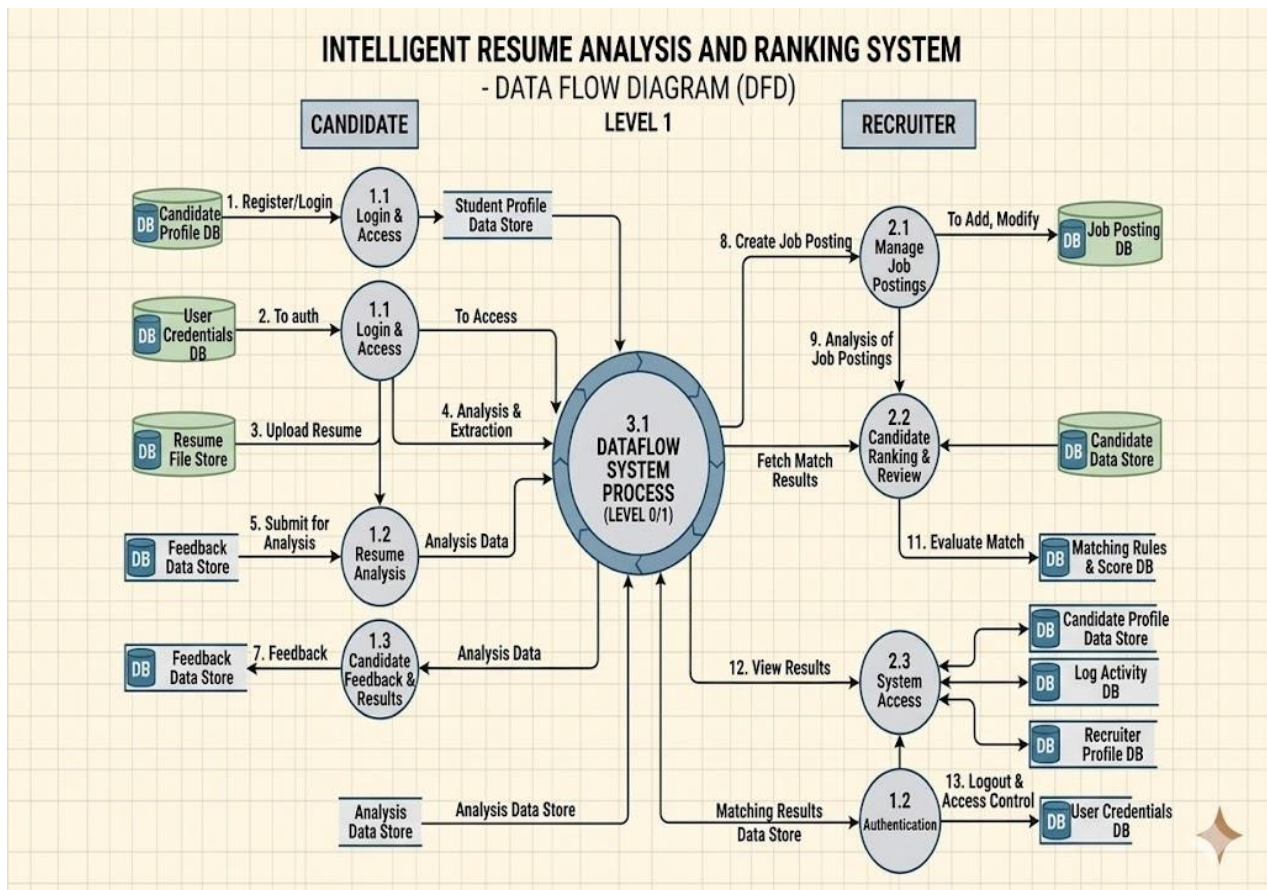


Fig.2 -Dataflow

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 Hardware & Software Specification

Hardware Specifications:

- **Development Machine:**
 - Processor Intel i3 or equivalent (recommended Intel i5 or higher for better performance)
 - RAM: Minimum 4 GB (8 GB recommended for smoother execution)
 - Storage: 256 GB HDD/SSD or higher for storing application files and database
 - Operating System: Windows, macOS, or Linux
- **User Devices:**
 - Desktop or laptop computers
 - Tablets and smartphones (iOS, Android)
 - Devices should support modern web browsers (Chrome, Firefox, Edge, Safari) for best experience
- **Network Requirements:**

Stable internet connection for accessing the online examination system and database

Software Specification

- **HTML5** and **CSS3** form the foundation for markup and layout, ensuring semantic structure and responsive design.
- **JavaScript** is used to handle user interactions, form validation, and dynamic content rendering.
- **Java EE** is used for backend development to manage system logic, processing, and secure data handling.
- **MySQL** is used as the database to store user details, resumes, job descriptions, and candidate evaluation results securely.
- **Apache Tomcat Server** is used for deploying and running the Java-based web application. Together, these technologies provide a reliable and efficient platform for conducting online examinations.

4.2 Proposed System Architecture

Setup & User Foundation:

Focuses on establishing the project architecture and fundamental user accounts. Deliverables include secure login and registration pages, user dashboard, and profile editing functionality.

Resume Processing & Analysis Module:

Focuses on implementing the main functionality of the system, which is resume upload and intelligent analysis. Deliverables include resume parsing, extraction of key details (skills, education, experience), and job description matching.

Evaluation & Ranking Engine:

Focuses on processing candidate data and generating results. Deliverables include similarity scoring, automated candidate ranking, and skill gap analysis for improved decision-making.

Administration & Final Delivery:

Focuses on empowering recruiters and administrators with management tools. Deliverables include a recruiter dashboard for job postings, candidate shortlisting, system monitoring, and final result visualization.

4.3 Domain

The Intelligent Resume Analysis and Ranking System operates across several important technological and recruitment domains, primarily focusing on automating the hiring process and improving candidate evaluation efficiency:

- **Human Resource Technology (HR Tech):** The primary domain. The system enhances the recruitment process by providing an intelligent platform for resume analysis, candidate evaluation, and hiring decision support.
- **Database Management Systems (DBMS):** Central to the application. The system relies on structured data storage using databases like MySQL to manage candidate profiles, resumes, job descriptions, and evaluation results efficiently.
- **Web Application Development:** The system is developed as a web-based application using technologies like Java, HTML, CSS, and JavaScript, enabling users (candidates and recruiters) to access the platform through web browsers.
- **Automated Evaluation Systems:** The platform uses intelligent matching techniques to analyze resumes and compare them with job requirements, reducing manual effort and improving ranking accuracy.
- **Data Analysis & Reporting:** The system provides insights such as candidate ranking, skill matching, and recruitment reports, helping recruiters make informed hiring decisions.

CHAPTER 5

RESULTS & CONCLUSION

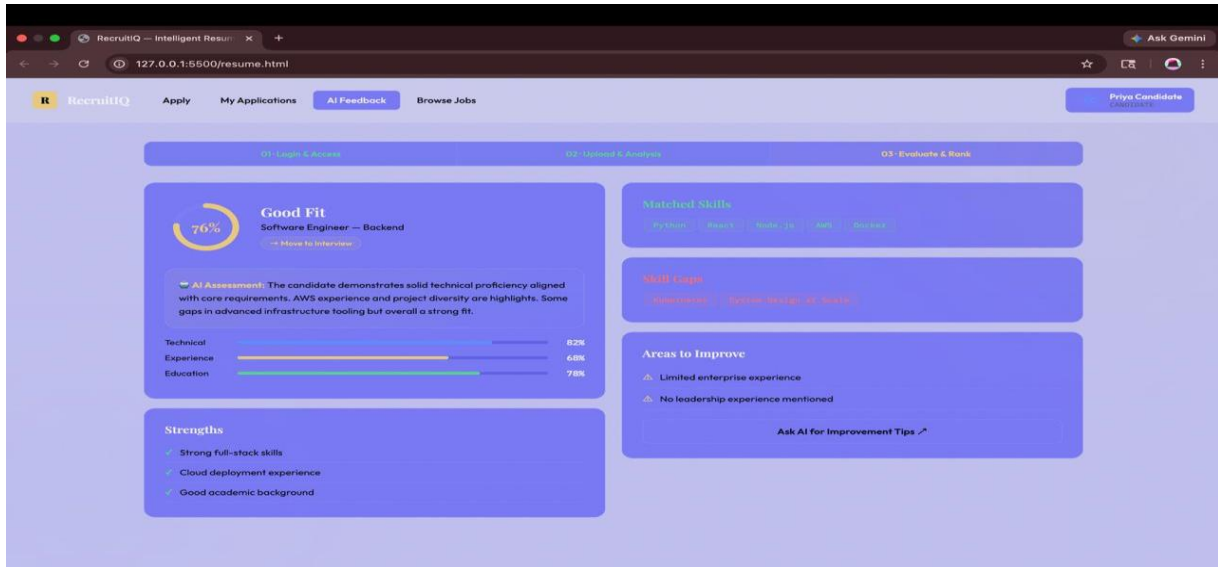


Fig.3 – Performance of Website

The success of the Intelligent Resume Analysis and Ranking System is measured through three key areas:

1. **Improved Recruitment Efficiency:** Measured by faster resume processing, reduced manual screening time, and quick identification of suitable candidates, leading to improved hiring speed.
2. **Enhanced Evaluation Accuracy:** Demonstrated by accurate matching of resumes with job descriptions, reduced human bias, and more consistent candidate ranking based on skills and relevance.
3. **Data-Driven Decision Making:** Evidenced by detailed reports and analytics such as candidate ranking, skill gap analysis, and recruitment insights that help recruiters make informed decisions.

The final result is the successful implementation of a unified platform that provides two key experiences: a user-friendly interface for candidates to upload resumes and track feedback, and an efficient recruiter dashboard for managing applications.

The Intelligent Resume Analysis and Ranking System effectively bridges the gap between traditional manual hiring methods and modern AI-based recruitment solutions. By automating resume evaluation and ensuring accuracy and fairness, this platform establishes a reliable and scalable solution for efficient and intelligent recruitment.

Future Enhancements

- **AI-Based Resume Screening System:** Implement AI-powered techniques to improve resume analysis by detecting skill relevance, experience depth, and job-fit accuracy. This will enhance the quality and reliability of candidate evaluation.
- **Advanced Analytics & Career Insights:** Enhance the system with detailed analytics to track candidate performance trends, skill distribution, and hiring outcomes. This can help recruiters make more informed and data-driven decisions.
- **Intelligent Job Matching System:** Introduce an adaptive matching mechanism where candidates are automatically recommended for jobs based on their skills, experience, and past applications, improving placement efficiency.
- **Mobile Application Support:** Develop a dedicated mobile application to allow candidates to upload resumes and track application status, while enabling recruiters to evaluate candidates anytime and anywhere.
- **Multilingual Support:** * Expand the system to support multiple languages to make it more accessible to users from diverse regions and improve overall usability.
- **Integration with Job Portals & LinkedIn:** Connect the system with external job platforms and professional networks to fetch real-time job listings and candidate profiles, creating a more comprehensive recruitment ecosystem.

APPENDIX

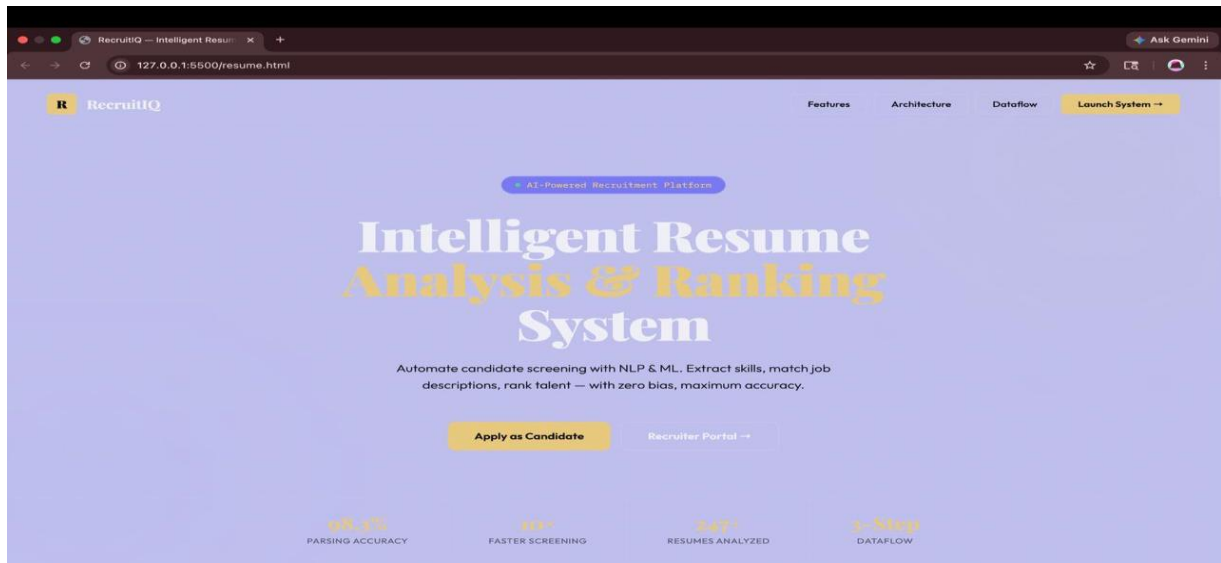


Fig.4: Home Page

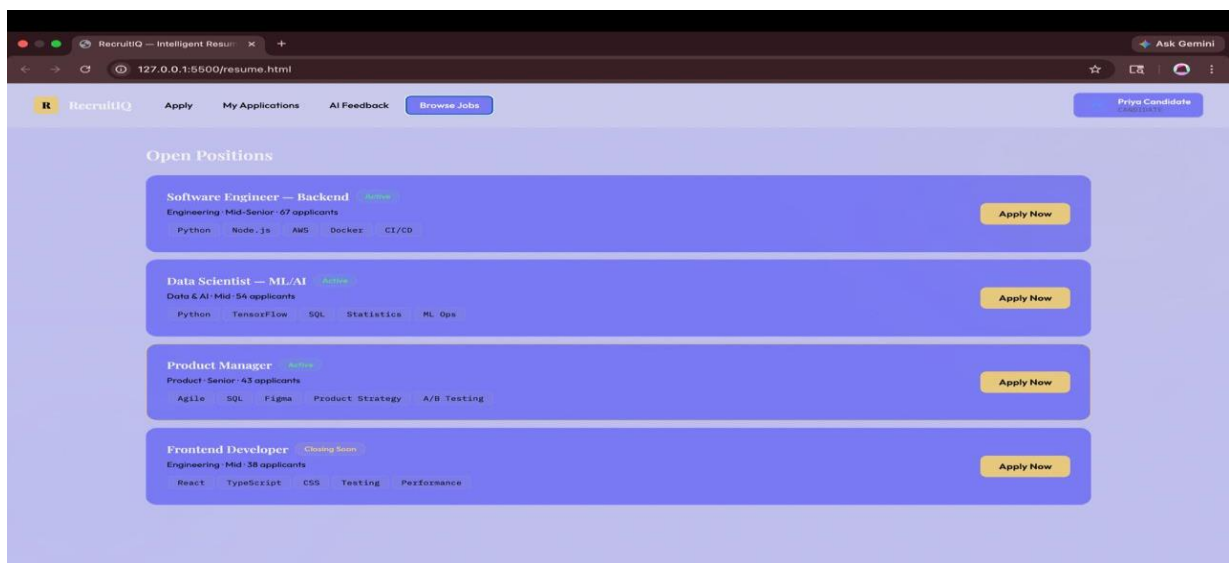


Fig.5: Candidate Browse Jobs

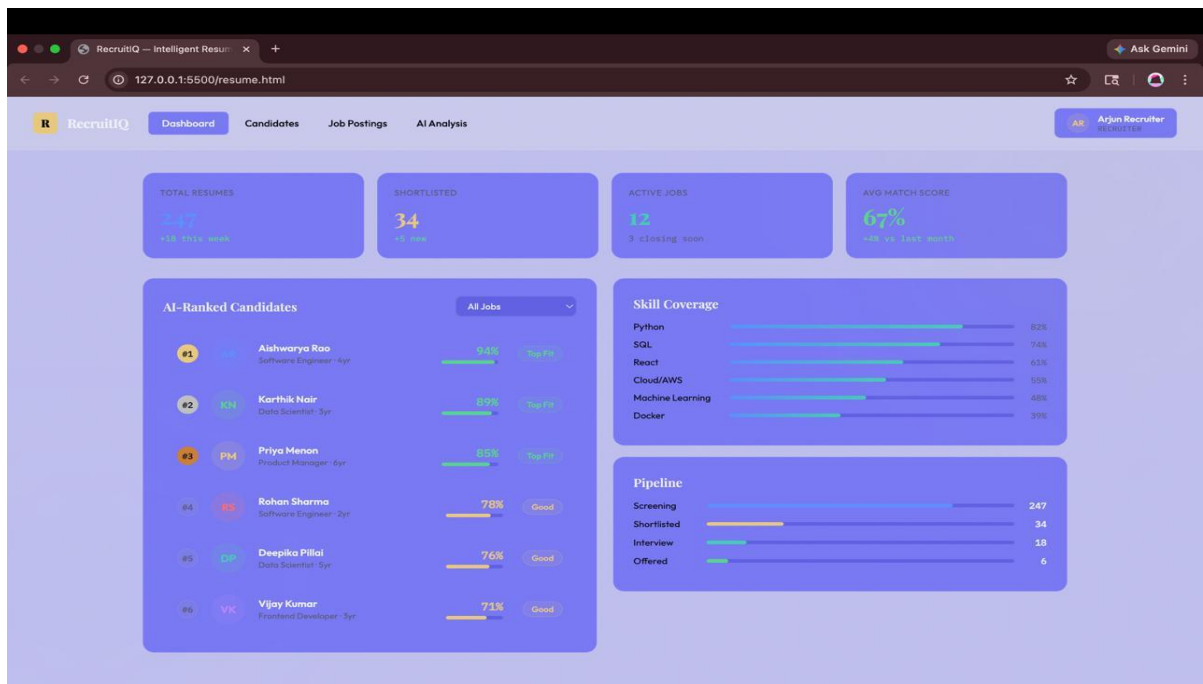


Fig.6: Recruiter Dashboard

All Candidates

	CANDIDATE	ROLE	EXPERIENCE	MATCH SCORE	STATUS	
#1	Aishwarya Rao a.rao@recruit.com	Software Engineer	4 years	94%	Shortlisted	View
#2	Karthik Nair k.nair@recruit.com	Data Scientist	3 years	89%	Under Review	View
#3	Priya Menon p.menon@recruit.com	Product Manager	6 years	85%	Interview Scheduled	View
#4	Rohan Sharma r.sharma@recruit.com	Software Engineer	2 years	78%	Applied	View
#5	Deepika Pillai d.pillai@recruit.com	Data Scientist	5 years	76%	Applied	View
#6	Vijay Kumar v.kumar@recruit.com	Frontend Developer	3 years	71%	Applied	View

Fig.7: Recruiter Candidates

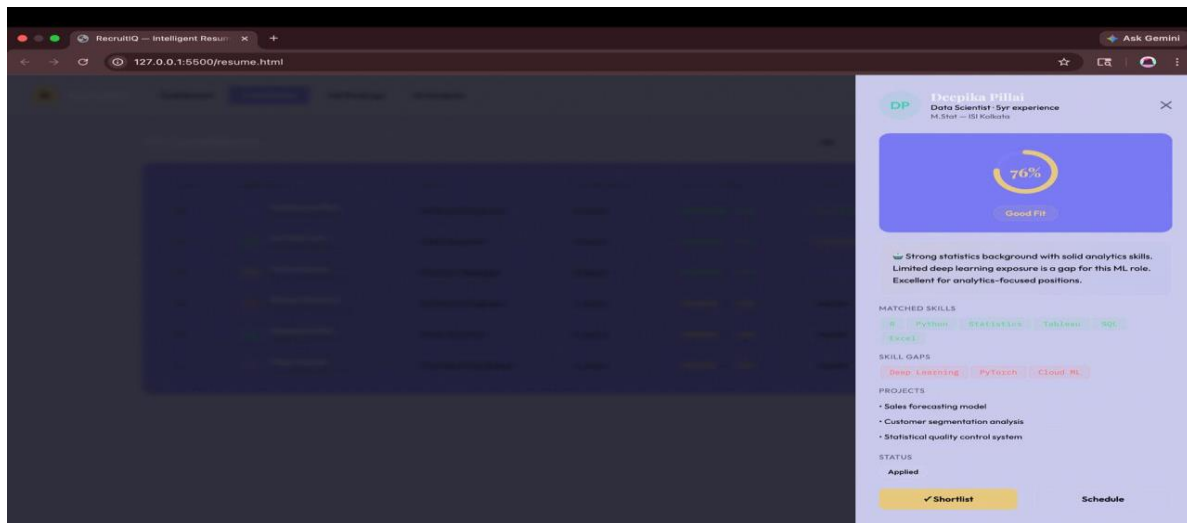


Fig.8: Recruiter Decision

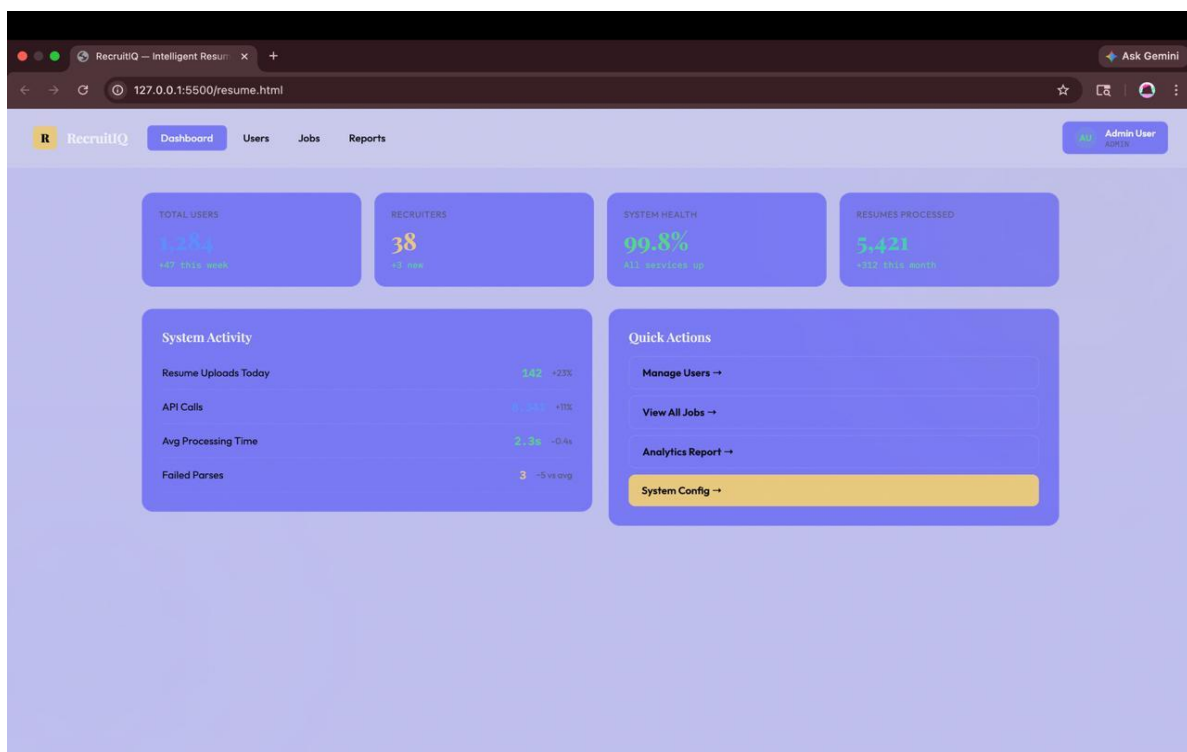


Fig.9: Admin Dashboard

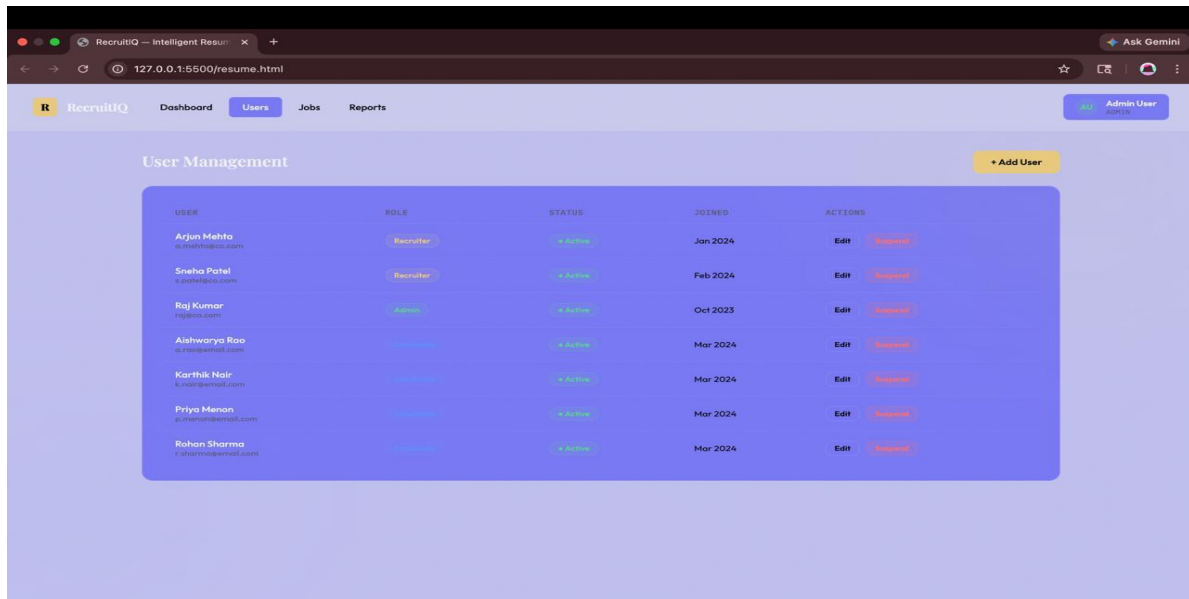


Fig.10: User Management

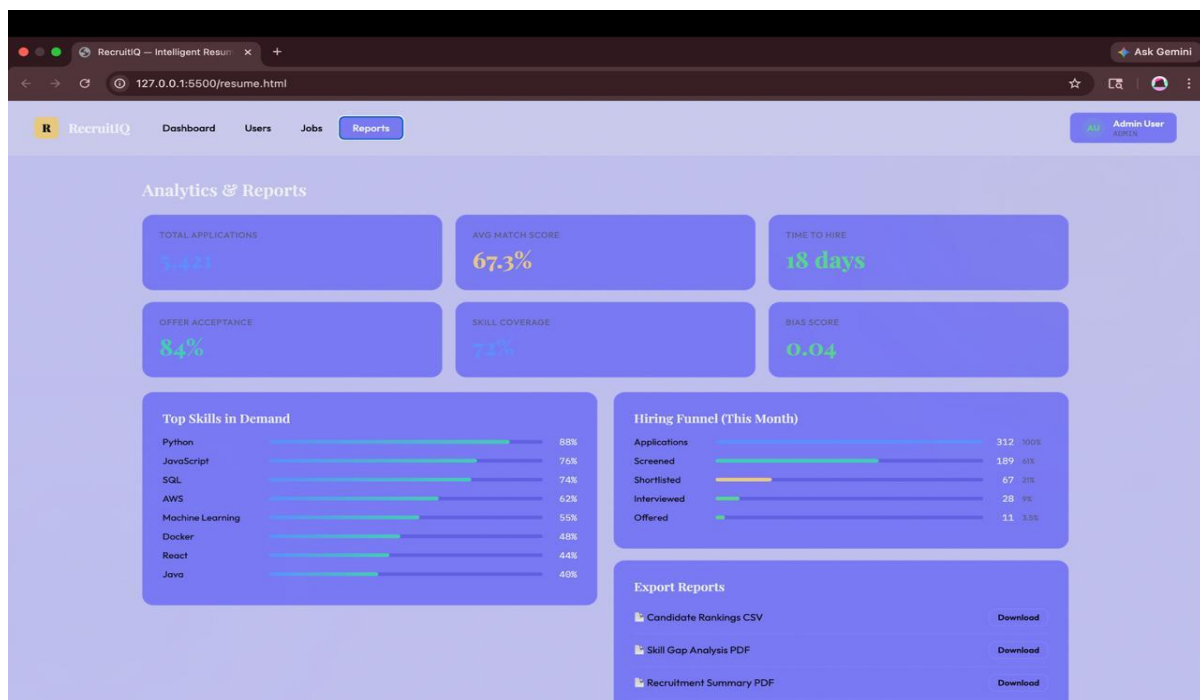


Fig.11: Admin Report